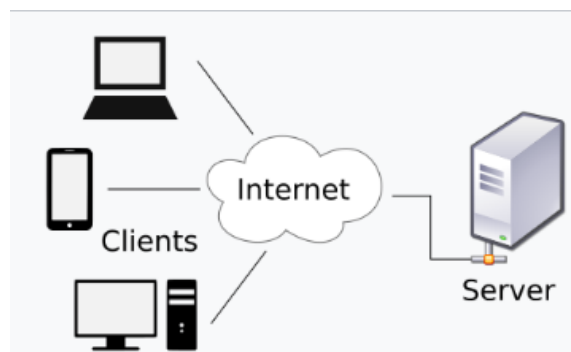


Client Server Architecture:

Client: A client is a computing entity (hardware or software) that requests services or resources from a server over a network. It acts as an interface for end-users or applications to access distributed services, often initiating communication.

Server: A server is a computing entity (hardware or software) that provides services, resources, or functionalities to one or more clients over a network. It listens for client requests, processes them, and returns the appropriate response. Servers can handle multiple client requests simultaneously and operate based on predefined protocols.



Peer To Peer:

The Peer-to-Peer (P2P) model is a decentralized network architecture in which each participating node, referred to as a peer, functions both as a client and a server. The P2P model allows direct communication and resource sharing among peers without a central authority. P2P networks can be classified into structured, unstructured, and hybrid models based on how peers discover and connect with each other.

Protocols

1. Network Communication Protocols:

enable data transmission between devices.

TCP	Transmission Control Protocol	Reliable, connection-oriented communication
UDP	User Datagram Protocol	Fast, connectionless communication
IP	Internet Protocol	Routes packets across networks
ICMP	Internet Control Message Protocol	Error reporting and diagnostics
ARP	Address Resolution Protocol	Maps IP addresses to MAC addresses

2. Application Layer Protocols: These protocols facilitate user services.

HTTP 80	Hypertext Transfer Protocol	Web communication
HTTPS 443	HTTP Secure	Secure web communication using SSL/TLS
FTP 20	File Transfer Protocol	File transfers between computers
SMTP-25	Simple Mail Transfer Protocol	Email sending
IMAP & POP3 110	Internet Message Access Protocol & Post Office Protocol 3	Email retrieval

3. Network Management Protocols

Used for monitoring and managing network devices.

SNMP 161/162	Simple Network Management Protocol	Network device monitoring
DHCP-67/68	Dynamic Host Configuration Protocol	Assigns IP addresses dynamically
NTP	Network Time Protocol	Synchronizes clocks across devices

4. Security Protocols: Ensure secure data transmission.

SSL/TLS	Secure Sockets Layer Transport Layer Security	Encrypts web traffic
SSH -22	Secure Shell	Secure remote login and file transfers

5. Data Link Layer Protocols: Control data transfer between directly connected devices.

Ethernet	(IEEE 802.3)	Wired LAN communication.
Wi-Fi	(IEEE 802.11)	Wireless LAN communication.
PPP	Point-to-Point Protocol	Direct communication between two nodes

LAN – Local Area Network: - small building/ office

MAN- Metropolitan Area Network: -city

WAN – Wide Area Network : - country

Modem: A modem (Modulator-Demodulator) converts signals between ISP and your network.

Router: A router manages and distributes internet connections within a network.

Topologies:

Mesh: - every device is connected to another device via a particular channel. Number of links required to connect NC2.

Bus: - Every computer and network device is connected to a single cable.

Star: - All the devices are connected to a single hub through a cable.

Ring: - it forms a ring connecting devices with exactly two neighbouring devices.

Network Devices

Access Point: - An access point in networking is a device that allows wireless devices to connect to a wired network.

Firewall : - A network security device that monitors and controls the flow of data between your computer and internet

Repeater: - To amplify the signal over the same network before the signal becomes too weak.

Hub: - A hub is a multiport repeater, connects multiple wires coming from different branches.

→ Hubs cannot filter data, so data packets are sent to all connected devices.

→ Active Hub : - It serves both as a repeater as well as a wiring center.

→ Passive Hub: -These are the hubs that collect wiring from nodes and power supply from the active hub. Cannot used to extend the distance between nodes.

→ Intelligent Hub: - works like active hub, enables administrators to monitor traffic passing through the hub

Bridge: - It operates at the data link layer. A bridge is a repeater, with add on the functionality of filtering content by reading the MAC addresses of the source and destination.

→ Transparent Bridges: - These are the bridge in which the stations are completely unaware of the bridge's existence

→ Source Routing Bridges: - routing operations is performed by the source station and the frame specifies which route to follow.

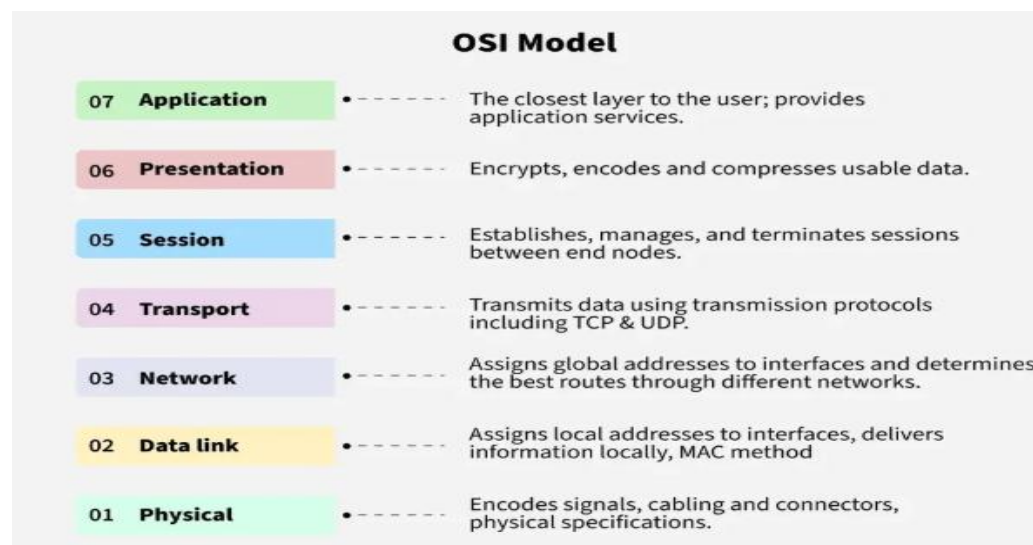
Switch : - A data link layer device, a multiport bridge with a buffer designed that can boost its efficiency(a large number of ports imply less traffic) and performance. It check errors, if errors present in the packet it doesn't forward the data further.

Gateway: - A passage to connect two networks, they work as messenger agents that take data from one system, interpret it and transfer it to another system. They also called as protocol converter

Transmission Modes:

- **Simplex Mode** : the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit, and the other can only receive.
- **Half-duplex Mode** : each station can both transmit and receive, but not at the same time.
- **Full-duplex Mode** : both stations can transmit and receive simultaneously.

OSI model:



Physical layer: -convert into bits and passed to data link and go to higher.

Transmission media: Physical path between sender and receiver

Guided media: twisted pair cable, coaxial cable, fibre optic cable

Unguided media: radio waves, microwave, infrared, satellite

Switching Techniques:

- Circuit switching: - Dedicated communication path is established for the entire communication session.
- Message Switching: - entire message is sent to an intermediate node, stored, and forwarded until it reaches the destination.
- Packet Switching: - data is divided into packets that are routed independently to the destination.

Data Link Layer

Flow Control Mechanisms : -

Stop and Wait Protocol : - Only one data packet can be shared over the link and the sender has to wait for the positive acknowledgment so that it can send another packet. It leads to poor efficiency and poor resource utilization.

Sliding Window Protocol: -

Go Back N : -known as “conservative protocol” and uses cumulative acknowledgments

Select Repeat ARQ: - retransmits only the lost or corrupted frames instead of all frames

Framing in DLL: provides for a sender to transmit a set of bits that are meaningful to the receiver. i.e, Character/Byte stuffing and Bit Stuffing

MAC address:

→ A MAC address is a unique identifier assigned to a network interface card (NIC) of a device for communication within a local network. It operates at the Data Link Layer (Layer 2) of the OSI model and is used for device identification in LANs.

→ They manage how devices share a common communication medium and ensure efficient and collision-free data transmission.

A) Connection Based Protocol: In these protocols, devices compete for the medium, and collisions may occur. They are suitable for networks with more traffic.

a. ALOHA:

i. Pure ALOHA transmits whatever they want.

ii. Slotted ALOHA transmission time is divided into slots.

b. CSMA (Carrier Sense Multiple Access)

i. Detects collisions and retransmits data.

B) Controlled Access Protocols: These protocols avoid collisions by controlling access to the medium.

Error Control

→ Hamming Code

→ Cyclic Redundancy Check

→ Check sum

Network Layer: Class Full Addressing is a method of dividing in five classes.

→ works for transmission of data segments that work for one computer to another computer in different network via ROUTER

Subnetting: dividing a bigger network into a smaller network.

→ A **subnet mask** is a 32-bit number used in IPv4 to divide an IP address into network and host portions. It helps identify the size of the network and enables subnetting by determining which part of the address belongs to the network and which to the host.

Supernetting: Multiple smaller networks are combined to form a bigger network.

CIDR (Classless Inter Domain Routing): - a method of IP address allocation and IP routing that allows for more efficient use of IP addresses.

→ an IP address of 192.168.1.0 with a prefix length of 24 would be represented as 192.168.1.0/24. This notation indicates that the first 24 bits of the IP address are the network prefix, and the remaining 8 bits are the host identifier.

Private IP: - used to communicate within the same network. Unique private IP Addresses are provided to each and every device that is present on the network.

Public IP : - used to communicate outside the network which it is assigned by ISP.

→Dynamic IP : - These IP addresses changes over time. ISP provides an IP address these random IP addresses are called Dynamic IP addresses

→Static: - Does not change with time. These are used by DNS servers.

Routing:

→determines how data packets reach their destination

→ Protocols :- RIP, OSPF, BGP

→Hop Count: - number of routers between the source and destination network, the path with lowest hop count is considered the best route to reach network.

→ Open Shortest Path First: - a link-state routing protocol that is used to find the best path between the source and the destination router using its own SPF algorithm.

→ Routing Information Protocol : - uses hop count as a routing metric to find the best path between the source and the destination network.

Router (Layer 3): Connects different networks, uses IP addresses.

Switch (Layer 2): Operates within a LAN, uses MAC addresses.

ARP: - Address resolution protocol request is a network protocol used to map an IP address to its corresponding MAC (Media Access Control) address within a local network.

Reverse ARP: - a network protocol used to obtain an IP address for a device (like a diskless workstation) from its MAC address.

Transport Layer: The TCP header is a fixed 20-byte (minimum) structure in the Transport Layer that includes fields such as source and destination ports, sequence and acknowledgment numbers, flags.

→ TCP Congestion Control: - manages network congestion by adjusting data transmission rate.

→ SYN → SYN+ACK → ACK

→ TCP Connection Termination: - process that ends TCP connection

- a) Graceful Connection release: it is opened until both parties have closed their sides of connection.
- b) Abrupt Connection release : - Either one TCP entity is forced to close the connection or one user closes both directions of data transfer.

→ 8-Byte structure includes source Port, Destination Port, Length and Check Sum.

Session layer: - responsible for establishing , maintaining and terminating sessions in a network for a certain time. It is responsible for identification of entities participating in a session.

Presentation Layer :- Translates the Data Packet from network layer to application layer. It responsible for protocol conversion, data encryption/decryption, character conversion and data compression.

→ takes data from application layer and convert into binary format

Application layer : - Directly interact with applications.

Application	HTTP, HTTPS, FTP, SMTP, POP3, IMAP, DNS, SNMP, Telnet
Presentation	SSL/TLS, JPEG, GIF, MPEG, ASCII, EBCDIC

Session	NetBIOS, RPC, PPTP, SOCKS
Transport	TCP, UDP, SCTP
Network	IP, ICMP, ARP, RIP, OSPF, BGP, IGMP
Data Link	Ethernet, PPP, MAC, VLAN, HDLC, Frame Relay
Physical	USB, Bluetooth, Ethernet (Physical), DSL, Wi-Fi, Fiber Optics

Encryption: - Encryption is the process of converting plaintext data into an unreadable format (ciphertext) using cryptographic algorithms to ensure confidentiality and data security.

→ AES(Advanced Encryption Standard) : - A symmetric encryption algorithm that uses fixed key sizes.

→ RSA (Rivest-Shamir-Adleman): An asymmetric encryption algorithm that uses a pair of public and private keys for secure communication and authentication

Attack: - A cyberattack is an intentional attempt to exploit vulnerabilities in computer systems, networks, or applications.

DDoS (Distributed Denial of Service) Attack: An attack that overwhelms a target server or network with excessive traffic from multiple sources, causing service disruption.

MITM (Man-in-the-Middle) Attack: An attack where an unauthorized entity intercepts and potentially alters communication between two parties to steal sensitive information or inject malicious content.

Network Address Translation: - it allows multiple devices to use same public IP address and access internet.

→ a process in which one or more local IP are translated into one or more global IP addresses.

Mask Port numbers: if two hosts have same destination of same port number, NAT changes the public ip address.

VPN: -A VPN is a secure communication technology that encrypts internet traffic and routes it through a remote server, ensuring privacy, anonymity, and data security by masking the user's IP address and protecting against cyber threats.

Bandwidth : - maximum data transfer rate of a network or internet connection, measured in bits per second (bps). It represents the capacity of a communication channel to transmit data over a given period.

Latency : - Latency is the time delay between the transmission and reception of a data packet in a network. It is typically measured in milliseconds (ms) and affects the responsiveness of online applications.

Throughput : - the actual rate at which data is successfully transmitted over a network, typically measured in bits per second (bps) or packets per second (pps).

Jitter: - Jitter refers to the variation in packet arrival times in a network. High jitter can cause disruptions in real-time applications like VoIP (Voice over IP) and video streaming

Ping : - Ping is a network utility that tests connectivity between two devices by sending ICMP (Internet Control Message Protocol) echo request packets and measuring the response time. It is used to assess network latency and packet loss.

Ipconfig : - ipconfig is a command-line utility in Windows that displays and manages the network configuration of a computer. It provides details such as IP address, subnet mask, default gateway, and can be used to refresh or reset network settings.

DNS Resolution : - The process of translating a domain name into an IP address using a hierarchical system of servers, including root servers, top-level domain (TLD) servers, and authoritative name servers.

SCTP (Stream Control Transmission Protocol): A transport layer protocol that provides reliable, message-oriented communication with multi-streaming and multi-homing capabilities, designed for applications requiring high availability.

Multiplexing: - A technique that combines multiple signals into a single transmission medium to optimize resource utilization, using methods like Frequency Division Multiplexing (FDM), Time Division Multiplexing (TDM), or Wavelength Division Multiplexing (WDM).

Token Passing : - A controlled access protocol in the Data Link Layer where a token circulates among nodes, granting permission to transmit data and preventing collisions in a network.

QoS (Quality of Service) : - A set of techniques to manage and prioritize network traffic to ensure performance metrics like low latency, high bandwidth, or minimal packet loss for critical applications.

SDN (Software-Defined Networking) : - A network architecture that separates the control plane (decision-making) from the data plane (forwarding), enabling centralized management and programmability of network devices via software.

NFV (Network Function Virtualization) : - A technology that virtualizes network functions (e.g., firewalls, routers) traditionally performed by hardware, running them as software on standard servers.

GSM/LTE/5G (Cellular Networks)

GSM (Global System for Mobile Communications): A 2G standard for digital cellular networks using TDMA for voice and SMS.

LTE (Long-Term Evolution): A 4G standard enhancing data speeds and capacity with IP-based communication.

5G: A next-generation standard offering ultra-low latency, high bandwidth, and massive device connectivity.

Traceroute : - A network diagnostic tool that traces the path of data packets from source to destination, identifying each hop (router) and measuring transit delays.

VLAN (Virtual Local Area Network) : - A logical segmentation of a physical LAN into multiple isolated broadcast domains, configured on switches to enhance security and reduce network congestion.

